The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Fall 2022

Mid Term Fall

Programming Fundamentals **CS-1101** Paper: Objective

Instructor: Engr.Muhammad Awais Marks: 20 Date: 02 –Feb -2023

Roll Number: _____ Time: 10 mints

Note:

- Encircle the label of correct option.
- Negative marking of **1** for each incorrect answer.
- Cutting is not allowed

#	**MCQ's (2 x 10 = 20)** **CLO1**
1	Which of the following is a correct identifier in C++? a. VAR_1234 b. $var_name c. 7VARNAME d. 7var_name
2	Identify the scope resolution operator? a. : b. :: c. ? d. None
3	Using which of the following data type can 19.54 be represented? a. void b. double c. int d. None
4	The constants in C++ are also known as? a. pre-processor b. literals c. const d. None
5	Identify the size of int datatype in C++? a. 1 byte b. 2 bytes c. 4 bytes d. Depends on compiler
6	C++ uses which approach? a. Right-left b. Top-down c. Left-right d. bottom-up
7	Identify the logical AND operator? a. || b. & c. && d. !
8	Find the output of: cout<< "10 / 0" << endl? a. Compilation error b. 0 c. Exception d. 10/0
9	Which operator has highest precedence in * / % a. * b. / c. % d. all have same precedence
10	What will be the output of the code? → `int main() { const int a=10; a++; cout<<a; return 0; }` b. 10 b. 11 c. Compilation error d. Run time error

The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Fall 2022

Mid Term

Programming Fundamentals **CS-1101** | Paper: Theory
Instructor: Engr.Muhammad Awais | Marks: 80 | Date: 02 –Feb -2023
Roll Number: _____ | Time: 1 hr 20 mints

Question 2: Solution

You are a software developer and you have been tasked to create a program that calculates the total cost of a car rental. The program should prompt the user for the type of car (economy, mid-size, or full-size), the number of days the car will be rented, and the number of miles driven. The cost of renting a car is $30 per day for an economy car, $40 per day for a mid-size car, and $50 per day for a full-size car. The cost of the rental also includes 10 cents per mile driven. The program should display the total cost of the rental.

a. **Write an algorithm for calculating the total cost of rental?**
   1. Start
   2. Prompt the user for the type of car (economy, mid-size, or full-size).
   3. Prompt the user for the number of days the car will be rented.
   4. Prompt the user for the number of miles driven.
   5. Based on the type of car, determine the daily rental rate.
   6. Calculate the total cost of the rental by multiplying the daily rental rate by the number of days and adding 10 cents per mile driven.
   7. Display the total cost of the rental.
   8. End
b. **Create a flowchart that shows the steps involved in calculating the total cost of the car rental?**
   It was simple so you can easily draw this.
c. **Write a C++ code for calculating the total cost of the car rental?**

```
#include <iostream>
#include <string>

using namespace std;

int main()
{
  string carType;
  int numDays, numMiles;
  double totalCost;

  cout << "Enter the type of car you would like to rent (economy, mid-size, full-size): ";
  cin >> carType;

  cout << "Enter the number of days you would like to rent the car: ";
  cin >> numDays;

  cout << "Enter the number of miles you will be driving: ";
  cin >> numMiles;
```

```
  if (carType == "economy")
  {
    totalCost = numDays * 30 + numMiles * 0.10;
  }
  else if (carType == "mid-size")
  {
    totalCost = numDays * 40 + numMiles * 0.10;
  }
  else if (carType == "full-size")
  {
    totalCost = numDays * 50 + numMiles * 0.10;
  }
  else
  {
    cout << "Invalid car type. Please enter either 'economy', 'mid-size', or 'full-size'." << endl;
    return 0;
  }

  cout << "The total cost of the rental is $" << totalCost << "." << endl;

  return 0;
}
```

Question 3 Solution

a.Identify the error in the following code and how can it be fixed?

```
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;
    cout << "Enter two numbers: ";
    cin >> num1 >> num2;

    if (num1 > num2)
    {
        cout << "The larger number is: " << num1 << endl;
    }
    else (num1 < num2)
    {
        cout << "The larger number is: " << num2 << endl;
    }

    return 0;
}
```

Else block do not have conditions. Remove this condition will fix the error

b. What is the output of the following code?

```
#include <iostream>
using namespace std;
int main()
{
    int x = 10, y = 5;
    cout << (x + y) * 2 - 53 / 2.0 <<endl;
    cout<<(x < y) && (x > 0) || (y < 20)<<endl;
    int result = y++ + y;
    cout << result <<endl;
    return 0;
`
```

3.5
0
11

c. What is meant by syntax and semantics in programming?

d. What is the purpose of the #include preprocessor directive in C++?

e. Differentiate between compiler, assembler and interpreter?

Question 4 (15+15) -CLO 2

Estimated Time : 35 mints

a. You are building a program for a fast food restaurant to calculate the total bill for a customer's order. If the customer orders a burger, the cost is $5. If the customer orders fries, the cost is $2. If the customer orders a drink, the cost is $1. If the customer orders all three items, they get a discount of $1. Write a code snippet in C++ that implements this logic.

```
#include <iostream>
using namespace std;

int main() {
    int burger = 5, fries = 2, drink = 1;
    int burger_order, fries_order, drink_order;
    int total;

    cout << "Enter the number of burgers ordered: ";
    cin >> burger_order;
    cout << "Enter the number of fries ordered: ";
    cin >> fries_order;
    cout << "Enter the number of drinks ordered: ";
    cin >> drink_order;

    total = burger_order * burger + fries_order * fries + drink_order * drink;
    if (burger_order && fries_order && drink_order) {
        total -= 1;
    }

    cout << "The total bill is: $" << total << endl;
    return 0;
}
```

b. A company wants to automate their employee salary raise process. They have the following rules for salary raise:

- If an employee has worked for less than 1 year, they are not eligible for a raise.
- If an employee has worked for 1-3 years, they are eligible for a 5% raise.
- If an employee has worked for 3-5 years, they are eligible for a 7% raise.
- If an employee has worked for 5 or more years, they are eligible for a 10% raise.

Using a switch statement, write a program that takes in the number of years an employee has worked at the company and outputs the corresponding salary raise percentage, or an error message if the employee is not eligible for a raise

```
#include <iostream>
using namespace std;

int main() {
    int years_worked;
    double raise_percentage;

    cout << "Enter the number of years worked: ";
    cin >> years_worked;

    switch (years_worked) {
        case 0:
            cout << "Error: Employee not eligible for a raise." << endl;
            break;
        case 1:
        case 2:
        case 3:
            raise_percentage = 0.05;
            break;
        case 4:
        case 5:
            raise_percentage = 0.07;
            break;
        default:
            raise_percentage = 0.1;
    }

    if (years_worked > 0) {
        cout << "Salary raise percentage: " << raise_percentage * 100 << "%" << endl;
    }

    return 0;
}
```